JYOTI KUMARI TIWARI

M.Tech Graduate | AI/ML Engineer | Python Developer

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PROFESSIONAL SUMMARY

Results-driven M.Tech graduate in Computer Science (AI/ML) from RGPV University with a strong academic background (B.Tech ECE, 9.23 CGPA, 92%). Experienced in building and deploying end-to-end machine learning solutions using Python, Scikit-learn, FastAPI, Flask, and Docker. Demonstrated ability to deliver production-ready ML APIs across diverse domains including healthcare, finance, and NLP. Currently strengthening Data Structures & Algorithms (DSA) skills with a target of 150+ problems. Seeking a software/ML engineering role with service-based companies in Hyderabad.

TECHNICAL SKILLS

Programming Languages: Python, C, C++

ML / Data Science: NumPy, Pandas, Scikit-learn, XGBoost, TF-IDF, Sentence Transformers, Random Forest, SVM, Logistic Regression

ML Concepts: Data Preprocessing, Feature Engineering, Model Training & Evaluation, Classification, Regression, RAG (Retrieval-Augmented Generation), NLP

Deployment & Tools: FastAPI, Flask, Docker, REST APIs, Git, GitHub, Jupyter Notebook

Currently Learning: Data Structures & Algorithms (DSA), Advanced Python, ML Theory & Practicals

PROJECTS

Medical AI Assistant with RAG | Python, LLMs, Semantic Search, FastAPI

- Built a RAG-based Medical AI Assistant leveraging LLMs and semantic search over a structured medical knowledge base
- Implemented FastAPI backend with source citations, safety monitoring, and evidence-backed response generation
- Integrated vector embeddings for semantic retrieval, improving answer relevance for medical queries

Fraud Detection MLOps System | Python, FastAPI, Docker, MLOps

- Designed end-to-end MLOps pipeline for fraud detection with model inference logging and failure debugging
- Containerized the full system using Docker; implemented monitoring dashboards for production-ready deployment

Job Skill Analyzer | Python, NLP, Sentence Transformers

- Developed AI-powered web app that compares resumes against job descriptions using NLP and sentence embeddings
- Generates skill gap analysis and a personalized learning roadmap for job seekers

House Price Prediction API | XGBoost, FastAPI, Docker

- Built end-to-end ML API using XGBoost on California housing dataset with full FastAPI and Docker deployment

Fake News Detection Web App | Flask, TF-IDF, Logistic Regression, Docker

- Developed a fully Dockerized Flask web application for fake news detection using TF-IDF vectorization and Logistic Regression

Other ML Projects (GitHub)

- Wine Quality Prediction: Random Forest model with Flask API deployment
- Rock vs Mine Prediction: Sonar signal classification with Docker setup
- Loan Status Prediction: SVM-based loan approval predictor via FastAPI
- Diabetes Prediction: End-to-end SVM + FastAPI + Docker ML pipeline

EDUCATION

M.Tech – Computer Science & Engineering (AI/ML)
RGPV University | 2023 – 2025 | 75%

B.Tech – Electronics & Communication Engineering (ECE)
RGPV University | 2019 – 2023 | 92% | CGPA: 9.23

Class XII – Science (Mathematics)
56%

Class X
66%

ONGOING DEVELOPMENT

- Actively solving DSA problems on coding platforms – target: 150+ problems (Arrays, Strings, Linked Lists, Recursion)
- Deepening Python fundamentals and ML theory with hands-on practicals (Iris dataset classification, data visualization)
- Revisiting core ML concepts: supervised/unsupervised learning, model evaluation metrics, cross-validation

GITHUB PORTFOLIO

9 public repositories showcasing end-to-end ML projects. View at: github.com/Jyotitiwari24